Muhammad Ibrahim

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Education

New York University May 2026

BSc General Engineering (Mechanical and Civil); UK 2:1 *Abu Dhabi, UAE*

- Coursework: Solid Mechanics, Engineering Dynamics, Fluid Mechanics, Thermodynamics, Heat Transport

New York University Jan 2024

Study Away, Immigration and Theatre *Paris, France*

Experience

Center for Stability, Instability and Turbulence (SITE) May 2026 – Jul 2026

Graduate Intern *New York, USA*

- Compiled a flow model's complete test dataset within six weeks, by operating the water-channel rig solo.
- Expanded the model's predictive range threefold, by grounding it in dimensional analysis.
- Eliminated an estimated 50+ tunnel runs, by training a Python and MATLAB model to predict new angles.

Advanced Manufacturing Workshop Sep 2025 – May 2026

Workshop Technician *Abu Dhabi, UAE*

- Lowered print failures from 50% to 10%, by diagnosing root causes and refining print settings.
- Strengthened parts 50% over plain filament, by reinforcing prints with carbon fibre and fibreglass.
- Fabricated load-bearing parts for three capstone teams on time, by running FEA and checking each print.

Center for Stability, Instability and Turbulence (SITE) Apr 2023 – May 2026

Undergraduate Research Assistant *Abu Dhabi, UAE*

- Engineered five water-tunnel models that showed the expected flow separation in every run, by co-designing them in SolidWorks for the machine shop and testing them with flow visualisation.
- Generated the centre's 130+ case fluid-instability dataset, by conducting each run with a research partner.
- Systematised every case for a follow-on study of laminar and turbulent flow, by jointly processing the results.

Projects

Advanced Anti-Roll Tank System (ART3S) — *SolidWorks, Simulink* Oct 2024 – Sep 2025

- Decreased simulated vessel roll **85%** passively and by **more than 98%** with active control, by designing 200 to 800 m^3 anti-roll tanks for ferry and car-carrier hulls in SolidWorks.
- Spearheaded design on an 8-person team that shortened roll decay time **33%** passively and **62%** under active control, by modelling the dynamics in Simulink and testing a 29 cm scale hull in a wave tank.

Traffic-Powered Savonius Wind Turbine — *ANSYS Fluent, Composites* Sep 2024 – May 2025

- Surpassed a non-serrated baseline by **72%** in peak moment and **56%** in power, by giving the blades serrated leading edges and a 45-degree helical twist in ANSYS Fluent CFD.
- Proved roadside self-starting within **2 to 5 seconds** and up to 10 V output, with 3 to 15 W projected at full scale, by building and testing a 1:3 scale carbon-fibre-reinforced rotor.

Extracurricular Activities

NYU Abu Dhabi Athletics, *Varsity Jiu-Jitsu, Recreational Sport and Team Videographer* 2022 – 2026

Habitat for Humanity International, *Engineer Volunteer, Dhulikhel, Nepal* 2025

Emirates Motorsports Organization, *Chief Track Marshal* 2019 – 2025

Skills

Design and CAD: SolidWorks (CSWA certified), mechanical design, system integration, rapid prototyping

Simulation: ANSYS Fluent (CFD), FEA, Simulink, control systems, dimensional analysis

Testing and Manufacturing: water tunnel and wave tank testing, flow visualisation, 3D printing, composites

Programming and Data: Python, MATLAB, machine learning, data processing, root cause analysis

Languages: English (native), Urdu (native), Arabic (elementary)